# Case II

## Stock-Out vs. Overstock:
## Fresh Bread Inventory at Save Supermarkets

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### Executive Summary

Fresh bread is Save Supermarkets' fastest-moving bakery item and one of our five best sellers, yet it is also a steady source of waste: over the last seven months the network threw away about 10.3% as many loaves as it sold. This report determines how much bread we should order, when we should order it, and what it would cost to raise in-stock performance from today's 80% service level to the managers' 90% goal and my long-term 95% goal.

The Economic Order Quantity (EOQ) model gives a cost-minimizing order of 12,852 loaves and an annual total inventory cost (TIC) of $575,089. That order would last about 6.6 days and would just fit in one 53-foot refrigerated trailer, but it is more than twice the bread's three-day shelf life. With a five-day lead time, the reorder point is 9,784 loaves.

Under a single-period (daily) policy, the safety stock needed is 225 loaves at 80%, 342 at 90% and 439 at 95%, for daily orders of 2,182, 2,299 and 2,396 loaves. Annual TIC at these quantities is $577,302, $577,143 and $577,024. A supplementary stock-out and disposal analysis shows the real cost of the extra bread rises by about $6,800 per year at 90% and $14,900 at 95%.

I recommend moving to 90% now with daily orders sized as expected demand plus safety stock, and reducing demand variability and disposal costs before pursuing 95%.

## 1. Introduction and Purpose

As the new Inventory Manager for Save Supermarkets, my first assignment is to review how our 25 stores replenish fresh bread. Bread is the fastest-moving product in our bakery and among our five best sellers, so running out disappoints customers who came in specifically to buy it, while ordering too much leads to spoiled loaves that cost $0.35 each to dispose of. Today we reach an 80% service level, meaning we have enough bread on about 80% of days; store managers want 90%.

This report is written for our Finance, Replenishment, Operations and Supply Chain teams. It (1) lays out the trade-offs of carrying more or less bread, (2) calculates the order quantity and reorder point that minimize cost under a continuous review system, (3) calculates the safety stock and order quantity needed for 80%, 90% and 95% service under a single-period system, (4) estimates the cost of each service level, and (5) recommends changes to how we manage bread inventory.

## 2. Methods and Analysis

### *Data*

Exhibit 1 records daily sales and disposal for 181 selling days (Monday to Saturday) over seven months. Following the case, each unit is treated as one loaf and daily demand is assumed to be normally distributed. Finance provided the remaining inputs: annual demand (D) of 700,000 loaves, a unit cost (U) of $0.82, an ordering cost (Co) of $10 per order, a five-day supplier lead time, 500 loaves per pallet and 26 pallets per 53-foot refrigerated trailer. Because bread spoils within three days, the case uses the disposal rate as the carrying-cost percentage (Ci). I computed each day's disposal as a percentage of that day's sales and averaged the 181 values, which gives Ci = 10.34%. Table 1 summarizes the data.

**Table 1. Summary of Exhibit 1 and case inputs**

| Measure | Value |
|---|---|
| Selling days observed | 181 (Mon to Sat, 7 months) |
| Average daily demand (μ) | 1,956.67 loaves |
| Standard deviation of daily demand (σ) | 266.78 loaves |
| Lowest / highest daily demand | 1,505 / 2,394 loaves |
| Average daily disposal | 198.94 loaves |
| Carrying cost % (Ci = avg. daily disposal ÷ sales) | 10.34% |
| Annual demand (D) / unit cost (U) / order cost (Co) | 700,000 / $0.82 / $10 |
| Lead time / shelf life | 5 days / 3 days fresh, ~1 month frozen |

### *Continuous review: EOQ, TIC and ROP*

A continuous review system tracks inventory constantly and places a fixed-size order whenever stock falls to a set level. Two tools define it:

- Economic Order Quantity (EOQ) is the order size that balances two opposing costs. Ordering often in small amounts drives up ordering costs; ordering rarely in large amounts drives up carrying (here, disposal) costs. EOQ = $\sqrt{(2 \cdot D \cdot Co / (U \cdot Ci))}$. Total Inventory Cost (TIC) adds the three annual costs: TIC = $Co \cdot (D/Q) + U \cdot Ci \cdot (Q/2) + D \cdot U$, which are ordering, carrying and purchase cost.
- Reorder Point (ROP) is the inventory level that triggers a new order so that it arrives just as stock runs out: ROP = average daily demand × lead time. As instructed, no safety stock is included in the ROP.

I also divided the EOQ by average daily demand to see how many days one order would last, and compared it with trailer capacity (26 pallets × 500 loaves = 13,000 loaves).

### *Single-period review: service level and safety stock*

Fresh bread is a classic single-period product: what is not sold within its short shelf life has no value. Under this approach we order each period's expected demand plus a safety stock buffer: $Q = \mu + z\sigma$, where $\mu$ is average daily demand, $\sigma$ is its standard deviation and z is the number of standard deviations needed to cover demand on a chosen share of days (the target service level). From the standard normal table, z = 0.842 for 80%, 1.282 for 90% and 1.645 for 95%. Safety stock is $z\sigma$.

### *Cost of each service level*

Following the case method, I substituted each service level's order quantity into the TIC equation. Because the TIC model does not directly capture the money lost to stock-outs or the bread thrown away each day, I also ran a supplementary expected-cost check using the case's $0.78 lost profit per loaf short and $0.35 disposal cost per loaf left over, based on 312 selling days per year (Appendix B).

## 3. Results

### *Q1. Trade-offs of carrying more or less bread*

**Table 2. Inventory trade-offs for fresh bread**

| | Reasons to carry MORE bread | Reasons to carry LESS bread |
|---|---|---|
| **Financial** | Avoids losing $0.78 profit per loaf short; fewer, larger orders cut the $10 ordering cost; full trailers lower freight cost per loaf; bread drives sales of other items in the basket. | Each unsold loaf costs $0.82 to buy plus $0.35 to dispose of; ties up cash in inventory; uses scarce refrigerated and frozen space needed by other products. |
| **Non-financial** | Higher service level and customer satisfaction; protects store traffic and loyalty for a top-five product; absorbs demand peaks and supplier delays. | Fresher product and better quality perception; less food waste, which supports our mission; less labor for rotation and disposal; simpler store operations. |

In short, more bread protects sales and customer loyalty, while less bread protects margin, freshness and our waste goals. The rest of this report puts numbers on that balance.

### *Q2 to Q5. Continuous review results*

**Table 3. Continuous review results**

| Item | Result |
|---|---|
| EOQ = √(2 × 700,000 × $10 ÷ ($0.82 × 10.34%)) | 12,852 loaves (≈1,286 cases) |
| Annual ordering cost = $10 × 700,000 ÷ 12,852 | $544.67 |
| Annual carrying cost = $0.82 × 10.34% × 12,852 ÷ 2 | $544.67 |
| Annual purchase cost = 700,000 × $0.82 | $574,000.00 |
| Total inventory cost (TIC) | $575,089.35 |
| Days one EOQ lasts = 12,852 ÷ 1,956.67 | ≈ 6.6 days (≈ 54 orders/year) |
| Trailer capacity = 26 pallets × 500 loaves | 13,000 loaves; EOQ = 25.7 pallets, fits (98.9%) |
| Reorder point = 1,956.67 × 5 days | 9,784 loaves |

The EOQ is 12,852 loaves (about 1,286 cases), and annual TIC is $575,089.35. At the EOQ, ordering and carrying costs are equal at $544.67 each, which is the defining feature of the optimum. Purchase cost ($574,000) is 99.8% of TIC and does not change with order size, so ordering policy only moves about $1,100 a year under this model.

At 1,956.67 loaves per day, one EOQ lasts about 6.6 days, and we would place about 54 orders per year. The EOQ equals 25.7 pallets, which fits in a single 26-pallet trailer at 98.9% of capacity. However, the trailer is not the binding limit. Fresh bread lasts three days and the frozen area holds only one pallet (500 loaves), so more than half of an EOQ-sized delivery would spoil before it could be sold. The reorder point is 9,784 loaves (1,956.67 × 5 days). Because our lead time is longer than the shelf life, any bread on hand when an order is placed would be too old to sell by the time it arrives.

### *Q6. Safety stock under a single-period system*

**Table 4. Single-period safety stock and order quantity (σ = 266.78, μ = 1,956.67)**

| Service level | z-value | Safety stock (zσ) | Order qty (μ + zσ) | Change vs. 80% |
|---|---|---|---|---|
| 80% (current) | 0.8416 | 225 loaves | 2,182 loaves | - |
| 90% (managers) | 1.2816 | 342 loaves | 2,299 loaves | +117 loaves |
| 95% (long term) | 1.6449 | 439 loaves | 2,396 loaves | +214 loaves |

Moving from 80% to 90% requires 117 more loaves of safety stock per day (+52%), and going to 95% requires another 97. Each step in service needs a bigger buffer than the last because we are covering rarer, higher-demand days. As a check, the model predicts about 254 leftover loaves per day at 80%; the actual average disposal was 199, so the model is a reasonable fit to our operation.

### *Q7. Annual cost of each service level*

**Table 5. Annual TIC at each service-level order quantity**

| Service level | Ordering cost | Carrying cost | Total (TIC) | vs. EOQ TIC |
|---|---|---|---|---|
| 80% | $3,209.25 | $92.44 | $577,301.69 | +$2,212.34 |
| 90% | $3,045.39 | $97.42 | $577,142.81 | +$2,053.46 |
| 95% | $2,922.17 | $101.52 | $577,023.70 | +$1,934.35 |

Under the TIC equation, all three daily policies cost about $1,900 to $2,200 more per year than the EOQ, mostly because ordering every day means roughly 320 orders a year instead of 54. Carrying cost rises with safety stock ($92 to $102), but TIC falls slightly as the service level increases because the larger daily orders reduce the number of orders. This shows a limit of the EOQ framework: carrying cost is charged only on the average inventory, so it hides the daily waste that comes with more safety stock.

The supplementary analysis (Table 6, Appendix B) captures that waste. Raising the service level from 80% to 90% cuts expected lost profit by about $4,200 but adds about $10,900 in disposal, a net increase of $6,769 per year. Going to 95% adds $14,865 per year compared with today. Purely on cost, the best service level is the critical ratio 0.78 / (0.78 + 0.35) = 69%, below where we are today.

## 4. Conclusions and Recommendations

The EOQ of 12,852 loaves minimizes cost on paper, but it ignores the main fact about bread: it spoils in three days. A perishable product must be ordered in quantities that can be sold while fresh, which makes the single-period model the right tool. Raising service levels costs real money, but it is a modest amount next to the $574,000 we spend on bread each year and the customer traffic that bread brings in. I recommend the following:

- Adopt the 90% service level now. Order daily (or for each delivery) using $Q = \mu + 1.28\sigma$, about 2,299 loaves at today's demand. The roughly $6,800 annual cost is about 1.2% of bread spend and is justified by fewer empty shelves for a top-five product that draws customers into the store. (Replenishment, Finance)
- Treat 95% as a long-term goal and reach it by reducing variability rather than adding bread. Safety stock grows in direct proportion to σ. Forecasting by day of week and by store with point-of-sale data could cut σ substantially, so we could reach 95% with today's buffer. (Replenishment, Supply Chain)
- Close the lead-time gap. A five-day lead time against a three-day shelf life forces us to order on forecasts we cannot correct. We should negotiate shorter or scheduled daily deliveries with our bakery supplier, or shift part of the volume to par-baked or frozen bread with expanded freezer space beyond one pallet. (Supply Chain, Operations)
- Lower the cost of overstock. Day-old markdowns, sales to discount channels and donations to food banks can turn a $0.35 disposal cost into partial recovery and support our mission of minimizing waste. Every cent saved on overage raises the cost-justified service level. (Operations, Finance)

- Measure and review weekly. Track in-stock rate, disposal percentage and forecast error by store, and update μ and σ monthly so order quantities follow demand. (Replenishment, Operations)

Together, these steps let us meet the managers' 90% goal now at a known cost, and create a path to 95% that improves both customer service and waste.

## Appendix A. Calculation details

- Carrying cost: Ci = average of (daily disposal ÷ daily sales) over 181 days = 10.34%; annual carrying cost per loaf = U × Ci = $0.82 × 0.1034 = $0.0848.
- EOQ = √(2 × 700,000 × 10 ÷ 0.0848) = √165,166,668 = 12,851.76 ≈ 12,852 loaves.
- TIC(EOQ) = 10 × 700,000 ÷ 12,851.76 + 0.0848 × 12,851.76 ÷ 2 + 700,000 × 0.82 = 544.67 + 544.67 + 574,000 = $575,089.35.
- Days of supply = 12,851.76 ÷ 1,956.67 = 6.57 days. Trailer: 12,852 ÷ 500 = 25.7 pallets ≤ 26 pallets (13,000 loaves).
- ROP = 1,956.67 × 5 = 9,783.34 ≈ 9,784 loaves (no safety stock).
- Single period: Q = 1,956.67 + z × 266.78, with z = NORM.S.INV(0.80) = 0.8416, NORM.S.INV(0.90) = 1.2816, NORM.S.INV(0.95) = 1.6449. σ is the sample standard deviation (STDEV.S) of the 181 daily values.
- TIC at service levels uses the unrounded Q (2,181.19, 2,298.56 and 2,395.48) in the same TIC equation as the EOQ.

## Appendix B. Supplementary stock-out and disposal model

For an order of Q = μ + zσ under normal demand, expected loaves short per day = σ·L(z), where L(z) = φ(z) − z·(1 − Φ(z)) is the standard normal loss function, and expected loaves left over = zσ + σ·L(z). Annual cost = 312 days × ($0.78 × loaves short + $0.35 × loaves left over). The cost-minimizing service level is the critical ratio Cu ÷ (Cu + Co) = 0.78 ÷ 1.13 = 69.0%. This model is a cross-check on the case's TIC method, not a replacement for it.

**Table 6. Supplementary expected annual stock-out and disposal cost (312 selling days)**

| Service level | Loaves short/day | Loaves left/day | Lost profit ($0.78) | Disposal ($0.35) | Total / change vs. 80% |
|---|---|---|---|---|---|
| 69% (cost optimum) | 53.0 | 185.5 | $12,908 | $20,262 | $33,169 / -$1,849 |
| 80% | 29.8 | 254.3 | $7,248 | $27,770 | $35,018 / - |
| 90% | 12.6 | 354.5 | $3,074 | $38,713 | $41,787 / +$6,769 |
| 95% | 5.6 | 444.4 | $1,356 | $48,527 | $49,883 / +$14,865 |